

Aya Burpee Bonus 5

```
module vending_tb();

    // generates random coin inputs to vending machine

    // until a randomly-chosen number of gumballs have been dispensed

    class coin_gen;

        // Q2: define class

        rand bit [2:0] num_gum;

        rand bit D, N;

        constraint c_num_gum {

            num_gum dist {3:=35, 4:=35, 5:=15, 6:=15};

        }

        function integer get_num_gum();

            return num_gum;

        endfunction

        function bit [1:0] next_coin();

            integer a = $urandom_range(0,2);

            case(a)

                0: begin D = 0; N = 0; end

                1: begin D = 0; N = 1; end

                2: begin D = 1; N = 0; end

            endcase

            return {D, N};

        endfunction

    endclass

endmodule
```

```

endfunction

endclass // coin_gen

parameter T = 20; // clock period
logic clk, Reset, D, N;
logic Ready, Coin, Dispense, Return;

vending dut (*); // vending machine module

coin_gen cg; // coin generator object handle

initial begin
    clk <= 0;
    forever #(T/2) clk <= ~clk;
end

integer i, var;

initial begin
    // Q3.1 instantiate and randomized ecoin_gen obj
    cg = new();

    // Q3.2 randomize coin_gen object
    cg.randomize();

    // if (!cg.randomize()) $finish;

    // first reset the system

```

```
Reset = 1; D = 0; N = 0; @(posedge clk);
```

```
Reset = 0; @(posedge clk);
```

```
// Q3.3
```

```
var = cg.get_num_gum();
```

```
for (i = 0; i < var; i++) begin
```

```
    while (!Dispense) begin
```

```
        {D, N} = cg.next_coin();
```

```
        @(posedge clk);
```

```
    end
```

```
    $display("dispensed at time %t", $time);
```

```
end
```

```
@(posedge clk);
```

```
$finish;
```

```
end
```

```
// Q4: define assertions
```

```
property within_five;
```

```
    @(posedge clk)
```

```
    Coin | => ##[0:3] Dispense ##1 Ready;
```

```
endproperty
```

```
assert property (within_five)
```

```
    else $error ("too many clock cycles");
```

```
endmodule // vending_tb
```